

Activity: "Test Investigator – Can You Explain Every Line?"

Duration: 25–30 minutes

Objective:

By the end of this activity, students should be able to explain **why every line of a MockMvc test is written**, not just copy it.

Instructions

Step 1 (5 minutes)

Put students into groups of **2–3**.

Give each group the **complete code** (the answer above).

Tell them:

"The code is correct. Your task is NOT to rewrite it. Your task is to become software investigators and explain WHY every important line is needed."

Step 2 (15 minutes) (All Groups)

Each group completes the worksheet below.

Code	Explain in Your Own Words
@WebMvcTest(StockController.class)	Why is this annotation needed?
@MockBean StockService	Why don't we use the real StockService?
when(stockService.getAllStocks()).thenReturn(List.of(hsbc));	What is Mockito doing here?
mockMvc.perform(get("/api/stocks"))	What does MockMvc simulate?
.andExpect(status().isOk())	Why is the expected status 200?
jsonPath("\$.symbol")	What does \$[0] mean?

Code

`Optional.empty()`

`header().exists("Location")`

`thenThrow(new IllegalArgumentException(...))`

Explain in Your Own Words

Why does it return 404?

Why should POST return a Location header?

Why do we use `thenThrow()` instead of `thenReturn()`?

Tell students:

Don't Google. Discuss with your teammates first.

Step 3 (10 minutes)

Choose different groups to explain one line each.

For example:

Group 1

Explain

`@MockBean`

`private StockService stockService;`

Expected explanation

Spring replaces the real `StockService` with a Mockito mock. This allows us to control what the service returns without calling the real business logic or database.

Group 2 and Group 5

Explain

`when(stockService.getStockById(99L))`

`.thenReturn(Optional.empty());`

Expected explanation

Mockito pretends that no stock with ID 99 exists. Therefore, the controller should return HTTP 404.

Group 3 and Group 6

Explain

```
mockMvc.perform(get("/api/stocks/99"))
```

Expected explanation

MockMvc simulates a browser sending a GET request to the controller without starting a real web server.

Group 4 and Group 7

Explain

```
.andExpect(status().isNotFound())
```

Expected explanation

Since the service returned `Optional.empty()`, the controller responds with HTTP 404 Not Found.

Challenge Question (Last 5 Minutes)

Ask them these questions:

1. Why do we use `when(...).thenReturn(...)` in some tests but `thenThrow(...)` in others?
2. Why does POST return **201 Created** while GET returns **200 OK**?
3. Why does `jsonPath("$.symbol")` use `$(0)`, but `jsonPath("$.companyName")` does not?
4. What would happen if we removed `@MockBean`?
5. Why do we use MockMvc instead of opening a browser?